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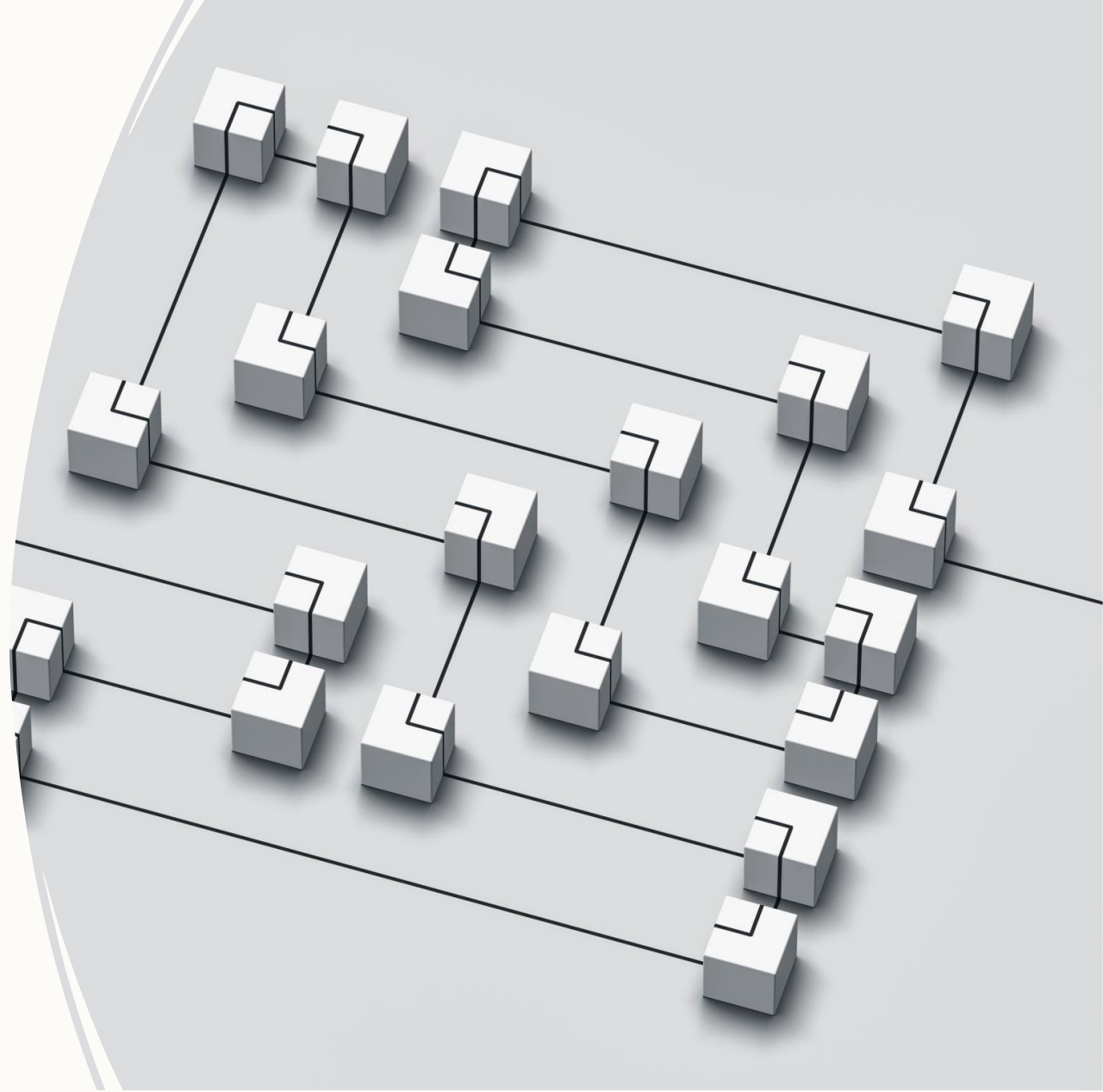
COMP 3056 Professional Ethics in Computing

Week 2 Theories of Ethics

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Learning outcomes

- What is Ethics, Morality and Law
- Some interesting observations about Ethics
- Five Ethical Theories:
 - Cultural relativism
 - Categorical imperative
 - Utilitarianism
 - Contractarianism
 - Ethics of caring
- Critical reasoning and Ethics





Definitions: Ethics, Morality and Law

- **Ethics**: Set of principles of how each member of a group ought to behave, shared by the group.
- **Morality**: One's individual belief about how one ought to behave. This typically implies an expectation of others.
- **Law**: Social rules of right and wrong, enforced by the state (i.e. police and law courts).

Difference between “is” (shi) and “ought” (yinggai):

- “Mr. Smith is paying his tax”. 事实
- “Mr. Smith ought to pay his tax”. 应当

关键区分：“is（事实）” vs “ought（应然）”——“他在纳税”“他应当纳税”。

Important note:

- These definitions are modern interpretations (from Reynolds and Brinkman & Sanders textbooks).
- Traditional philosophy: **ethics is philosophy of morality.**
- **Please remember this when self-reading on this topic.**



Relations: Ethics, Morality and Law

- These three ideas overlap, but are not necessarily the same. 有重叠但不完全相同



Examples:

Mr. Smith pays his taxes.

Moral ☒ Ethical ☒ Legal ☒

Lucy sells heroin to teenagers.

Moral ☐ Ethical ☐ Legal ☐

John is a gangster belonging to a criminal gang (ethical group). He withholds information about other gang members under interrogation by police.

Moral ☒ Ethical ☒ Legal ☐

It depends! : this is only true from John and the gang's perspective!

BigMoney is an online marketing company targetting elderly computer users, selling medical support products. They charge double the market price.

Moral (depends) Ethical ☐ Legal ☒



Some interesting ideas about Ethics

- Ethical decisions may differ between different groups
- Context
- “Cost” is important
- Egoism versus altruism
- Framework for Ethical Decision-making



Ethical decisions may differ between different groups

伦理因群体而异

- Ethics depends on the group which is being considered.
 - Example 1: In many Muslim countries, selling and drinking alcohol is unethical.
 - Example 2: In 18th century, slavery was ethically acceptable in many countries, but now is not.
 - Example 3: It is ethical to use marijuana in California, but not Texas; also, different view held by younger and elder people.
- Even between two rational people in the same culture, they may make different **ethical decisions**.
 - Example: If a close elderly family member is very ill, in great pain, and unable to look after themselves, is it ethical to seek humane assisted death?
- This is important, since it is clear there are different viewpoints.
- For this module: **listen carefully, respect other viewpoints, do not personalize discussions.**



Context 情节

- Ethical decisions often depend on context.

Example:

- Mrs. Smith lies to police that she is harboring people wanted by the police in her home.
- *Is this wrong or right?*
- Suppose additionally: Mrs. Smith is living in Nazi-occupied Denmark in 1942 and the people wanted by police are a Jewish family who have committed no crime.
- *Is she right or wrong?*

Lessons:

- Sometimes an ethical decision cannot be made until more information is gathered.
- Sometimes a poor ethical decision is made on misinformation.
- Sometimes an ethical decision needs to be changed in the light of new information.

没有情境，就没有可靠的伦理判断；有了关键情境，结论常会翻转。



“Cost” is important

代价 (cost) ”或“ 后果 (consequence) ” 的重要性

- The consequence or “cost” of ethical decisions is important.

Examples: which of these is more severe? 伦理判断不是简单的“对/错”二元问题，而是有“程度 (degree)” 的差异。

- A programmer turns up 10 minutes late to a software development meeting.
- A programmer is drunk at work.
- A software engineer deliberately fakes outcome of some tests on control software for car brakes.

Lessons:

不同行为有不同程度的不道德性 (unethical degrees)

- Actions have different unethical “degrees” (not just true/false).
- Should we (can we) “cost” outcomes quantitatively?



Egoism versus altruism

利己主义 (Egoism) vs 利他主义 (Altruism)

Two distinct philosophical viewpoints:

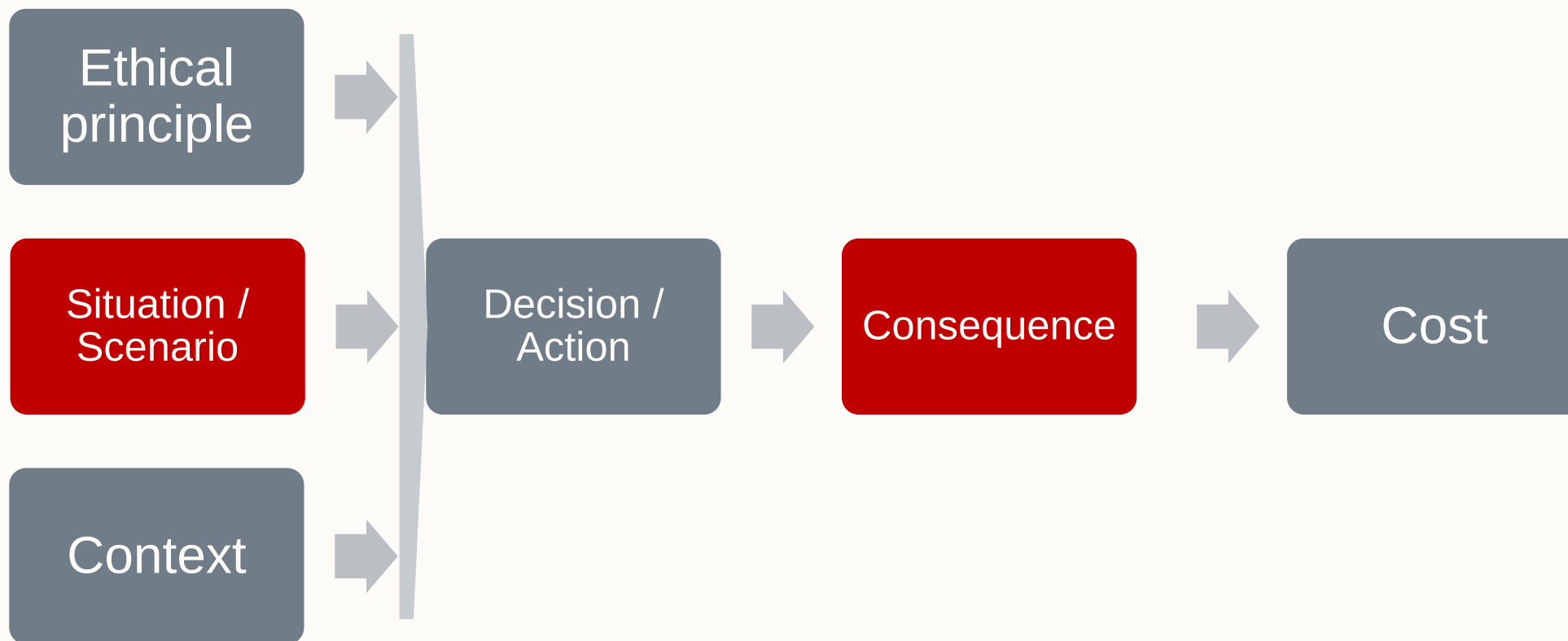
- **Egoism:** 人做任何事情都是为了自身利益 (self-interest) 极端利己主义不具可持续性，因为它破坏社会互惠的基础。
 - Everyone is motivated to do only what is good for themselves as individuals.
 - Hence, only ethical theory is “Do what is best for me”. “唯一的道德原则就是——做对我最有利的事 (Do what is best for me)。”
 - Clearly, this is problematic: Selfish and uncaring society. 自私，冷漠
- **Altruism:** 道德行为应以他人福祉为中心。
 - Ethical principle: Do what is good for other people.
 - Altruism may not be realistic!
 - At its extreme, this is not good because people may not care for themselves. 走向极端时，人可能忽略自我需求，导致被剥削或身心耗竭。
- **Modern approach blends egoism and altruism:** 遵守社会规则 (altruism) 反而有利于个人 (egoism)
 - 1. Following social rules (altruism) is good for individual (egoism) 照顾好自己 (egoism) 是帮助他人 (altruism) 的前提
 - 2. Caring for self (egoism) is a requirement before helping others (altruism).
 - Q: Can you think of an example of (1) and (2)?
 - (1) 例如：程序员遵守开源许可条款，不仅帮助社区，也能获得专业认可与就业机会。
 - (2) 例如：医生需要保持健康才能照顾病人；开发者需先平衡工作与休息，才能产出高质量代码。



Framework for Ethical Decision-making

伦理决策框架

To help us discuss ethical decision-making we can therefore provide the following general framework:





Theories of Ethics

- Cultural relativism
- Categorical imperative
- Utilitarianism
- Contractarianism
- Ethics of caring



Theories of Ethics

- The quest for defining ethics is an attempt to answer these questions:
 1. How do I know that X is good?
 2. Why is X good?
- We will be looking at ways to answer these questions.
- Theories of ethics developed by philosophers for millennia.
- Large and complex topic. In this lecture we only look briefly at five theories which have modern impact.
- Important note: Focus is on Western philosophy, following the textbooks, but many important Eastern philosophers have written on ethics.



1. Cultural Relativism

文化相对主义

- Ethical principle:

一个行为的“好”或“坏”，只能根据它所在社会的标准来判断。

An action is judged good or bad based only on the standards adopted by one's society.

- This follows from observation that ethics is different between groups. 同群体之间的伦理标准确实不同
- No valid *rational* criterion for determining the right thing to do exists. 没有一个普遍理性的标准能判断什么行为“绝对正确”
- Different cultures cannot be judged against each other 不同文化不能互相用各自的伦理标准来批判

- **Positives:** Promotes tolerance 包容
 - We do not unfairly judge other cultures

- **Negatives:** Difficult to defend 难以自洽
 - (e.g., slavery in 18th century);

- Questions:
 - What are the boundaries of a culture?
 - How do we determine whether two people belong to the same culture?

这些问题揭示了文化相对主义的模糊性与局限性。



Do Maasai and Canadian have same ethics?



1. Cultural Relativism (Pluralism) 多元主义

文化相对主义的问题在于——它缺乏客观性 (objectivity)

- Cultural relativism is problematic because it is not objective: we cannot apply the same ethical standards to people from different cultures.
- We need a *universal ethics code* for the *computing profession*.

Pluralism:

- However, two attributes of culture appear to be universal amongst all humans: *language and morality, there is a “natural morality”*:

The existence of a common morality is supported by the widespread agreement of most moral matters by all moral agents... killing, causing pain or disability, depriving of freedom ... unless there is an adequate justification ... deceiving, breaking promises, cheating, breaking the law, and neglecting duties also need justification in order not to be immoral.

[Bernard Gert , Common Morality , 2004, pp.8-9]

However, for us on this module, the case for cultural relativism is weak

– It does not provide clear guidance in deciding what is right when interacting with people of other cultures.

“For us on this module, the case for cultural relativism is weak.”

原因:

1. 它不能清楚指导我们如何在跨文化情境中做正确决策;
2. 它过于模糊, 无法应对全球性职业伦理标准的需要 (例如 AI、公平性、隐私等问题);
3. 相比之下, Pluralism 提供了更可行的折中立场: 承认多样性, 同时维护人类共同道德底线。



2. Categorical imperative 绝对命令

- Ethical principle:

如果某个行为能被当作**普遍化规则 (universal rule) **而不会自相矛盾，那么它就是道德的。

An action is moral if it can be stated as a universal rule without contradiction.

- Example:

- Suppose “It is OK for me to steal”.
 - This presupposes that people own personal property (that can be stolen)
 - State “It is OK for me to steal” as a universal rule: “it is OK for *anyone* to take *anything*”.
 - But if it is admissible for anyone to take anything, then the whole idea of personal property is meaningless.
- Therefore:-
 - There is no such thing as stealing, or,
 - Stealing is wrong.

- This derives ethics directly from **reason**.
- The process is called the “*categorical imperative*” and was introduced by German philosopher Immanuel Kant (1724 1804).

1 前提假设 (presuppose)

- 要“偷”，必须承认别人拥有私人财产；
- 否则“偷窃”这个概念本身就不存在。

2 将其普遍化 (universalize)

- 把这条规则扩展为：“任何人偷窃都可以”。

3 出现矛盾 (contradiction)

- 如果人人都可以偷，那么财产的概念将不复存在；
- 也就是说，若人人都偷，便无物可偷；
- 因此“偷窃”这条规则一旦普遍化，就自我否定。

4 结论 (Therefore)

- 因为该行为无法普遍成立，所以是不道德的；
- Stealing is wrong.





2. Categorical imperative

- A flaw with the categorical imperative is that it forms universal laws that must be logically true.
- No room for exceptions.
- Consider this scenario:
 - Yifan starts works at a software helpdesk. He is horrified to learn the company is tricking customers to buy additional security software that they do not need and does not even work. He know where documents to prove this are kept in the office building. Unseen, he takes them and hands them over to the police authorities.

But “taking” the documents is “stealing”, and the categorical imperative says he is wrong.

But is he? What do you think?



Deontological Ethics

义务论伦理学

- Categorical imperative is a type of Deontological Ethics:

道德判断关注“人做了什么（what people do）”，而不是“行为导致了什么结果（consequence）”。

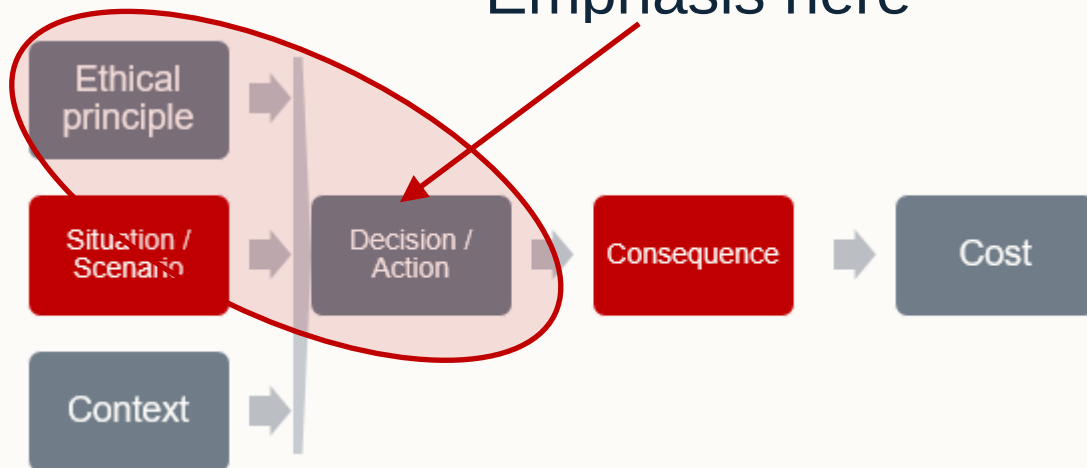
- Concerned with **what people do**, not with consequences of actions.
- Do the right thing because it is your duty.

义务论 = 按道德规则行事，而非凭结果决定。

- “Deon” is Greek word for duty.

表示义务论强调的是“行为与原则”部分，而非“后果”。

Emphasis here





Good intentions

- Deontological Ethics focusses on the **intention of an action.** 行为的动机与意图
- Principle of “good intention”.
- But if an ethical decision is **badly informed** then it may lead to **bad consequences.** 如果信息不足或判断错误，义务论只看“意图”，不看“结果”，因此可能忽略现实风险。

- Example:

Bob has to go home, but wants to continue working. Therefore he emails the files he was working on to his personal account. However, the files he was working on including personal data, were stolen (external email is not secure), and he caused a data breach.

- So Bob was well-intentioned, but the outcome was bad.

◆ 从义务论视角看：

- Bob 的意图 (intention)：继续工作，提高效率 → ✓ “好意”
- Bob 的行动 (action)：将文件发到私人邮箱 → 履行“工作义务”，但违反数据安全规定
- Bob 的结果 (consequence)：数据泄露 → ✗ 后果严重

义务论判断：

Bob 的行为出于善意，因此是道德的；

结果的好坏不影响行为的道德性。



3. Utilitarianism 功利主义

享乐主义

- **Hedonism** is the principle that actions that **make me happy are the best**. However, this is **egoistic**. 自我的
- **Utilitarianism** is the altruistic development of hedonism: 功利主义是享乐主义的“利他主义 (altruistic)”发展。

最正确的决定，是能让所有受影响的人获得最大幸福的那个决定。

The right decision is the one that causes the most happiness, amongst all people affected by the decision.

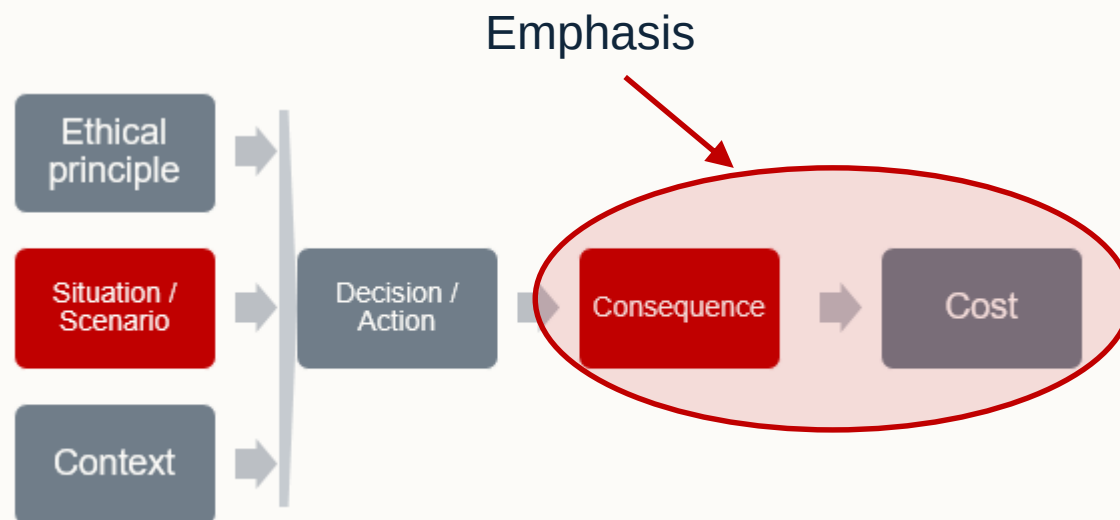
Jeremy Bentham (1748 1832) regarded as founder;
John Stuart Mill (1806 1873) regarded as strongest proponent.

- Utilitarianism is a **reaction to Deontological Ethics** focusing on the **consequence**.
- Hence, **it is a form of Consequentialism**: focus on **outcome**, not **intention**.

功利主义看后果而非意图



3. Utilitarianism



Maximize happiness (consequence)

- But what is “happiness”?
Physical, emotional, mental:
 - Satisfaction of accomplishment, appreciation of interesting idea, aesthetic pleasure of art, pleasure of helping others, ...
 - Absence from pain and suffering.
- Sometimes “happiness” is called “utility” (usefulness).



3. Utilitarianism



Example:

You can spend \$1,000 on educational resources for children. You have two options:

1. Spend \$1,000 on books for local library.
 2. Spend \$1,000 on e-books for distribution on the school intranet.
- One physical book costs \$10 and e-books are \$8 on average.
 - 95% of children attend the library each week, whereas only 70% look at the school intranet at home each week.

Therefore,

- Average number of library books accessible to each child = $0.95 \times 1,000 / 10 = 95$.
- Average number of e-books accessible to each child = $0.7 \times 1,000 / 8 = 87.5$.

Hence, measuring happiness by access to books, option 1 is best.



3. Utilitarianism

功利主义的挑战在于：虽然“最大幸福”听起来简单，但“谁的幸福、怎么算、何时算”却极难回答。

- Utilitarianism has been very influential in Economics. 功利主义在经济学中极具影响力
- However, utilitarianism requires a way to quantify or “cost” the consequence.
 - This is not very easy and may lead to crude oversimplification.
 - It may not even be possible.
 - For example, how to compare the pleasure of eating an apple and a banana? Is this the same for all people?
 - A utilitarian calculation is required by ordinary people for all actions?
 - This is called *act utilitarianism*.
 - An alternative is called *rule utilitarianism* which provides general rules that have already been costed (on average!).
- Utilitarianism either can only be applied after the decision (when cost is known), or using a *prediction* of consequence and cost, which may be wrong / misinformed.



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3. Utilitarianism

- Utilitarianism has a worse problem: “tyranny of the majority”. 多数暴政
- Because utilitarianism seeks to maximize total happiness amongst all affected people, then improving happiness at the expense of a minority is ethically acceptable, according to utilitarianism. 如果让多数人更快乐的方式是牺牲少数人的幸福或权益，功利主义仍然认为“合理”
- Consider the example on slide 21 again:
We chose to invest in books for the library. However, the 5% of children who do not attend, may not be able to because of mobility problems (disability or lack of transport resource). This decision would exclude them from access to the books.
- Of course, the “utilitarian calculation” can be adjusted, but this is ad hoc, and relies on introduction of new principles.

“功利性计算”是可以调整的，但这是临时的，并且需要引入新的原则。



3. Utilitarianism

Consider these questions from the **perspective of utilitarianism:**

1. Should we give all school children candy for lunch?
Consider short and long term effects.

短期结果 (short-term effects) 孩子们会很开心 “即时幸福” 增加；符合功利主义的“最大化快乐”原则。

长期结果 (long-term effects) 糖分过多导致健康问题（肥胖、蛀牙、糖尿病）；学习注意力下降；医疗成本上升，家长担忧增加。

结论：

功利主义要求我们综合考虑**总体幸福的总和**，包括**短期和长期后果**。

尽管短期快乐增加，但长期痛苦更大 功利主义会认为“**不应该给糖**”。

2. A profoundly unhappy person has no friends or family. Is it morally right to kill this person?

按纯粹功利主义逻辑：如果他的死亡能**减少痛苦总量**、提升“总体幸福”（通过消除痛苦），那么“杀死他”似乎在逻辑上是“道德的”。

但这引发严重的伦理问题：它**忽视了生命本身的尊严 (inherent value of life)**；可能导致**滥用与误用**（谁来判断“谁不值得活”？）；会破坏社会对生命权的基本信任（即长远看总体幸福反而下降）。



4. Contractarianism

- Both a political and a moral theory. 政治和道德理论
 - Initially from writings of English philosopher Thomas Hobbes (1588 1679): *The Leviathan*.
- Can be merged with consequentialist , or deontological theories.
- Premise: 前提
 - People are rational and wish to promote self interests (egoism). 人是理性的，会追求自身利益。
 - The best way of doing that is to promote the common interests of society.

但要实现个人利益，最有效的方式是促进社会共同利益。
- Ethical decisions are those that are in accordance with common interests; the social contract. 道德行为是那些符合共同利益、维护社会契约的行为





4. Contractarianism

Social Contract: 社会契约

- Originally, we were in a state of nature:
Humans live “*solitary, poor, nasty, brutish, and short.*” [Hobbes]
- The only way to overcome this “state of nature” is to come together with others for mutual benefit.
- We enter into contracts in which each party agrees to certain obligations in exchange for other benefits
- Contracts are often implicit i.e., not written down, or spoken about); but they are enforced by social norms and state institutions (police, law courts, government).



5. Ethics of Care 关怀伦理

- Originated by Nel Noddings, American feminist and philosopher [1929-2022]
- Critical of Utilitarian and Kantian approaches.
- Believes logic and maths not helpful for moral philosophy:
“The hand that steadied us as we learned to ride our first bicycle did not provide propositional knowledge”. [Noddings]
- A single moral ideal virtue : **caring for other individuals**.
- Goodness is whether or not we take care of people around us.
- **One-caring: someone who fulfils the role of a care giver.**
- To act morally, we need to make ourselves one caring for those with whom we have relations. 道德行动不是遵守外在规则，而是在关系中付出关怀



Components of Caring

真正的关怀不仅是表面的帮助，而是一种全神贯注地体察他人需要的情感参与；

“关怀”包含两个层面：

- 1 情感上的投入（感同身受、理解他人）；
- 2 行动上的回应（实际照顾或帮助对方）。

专注投入

- Caring is characterised by feeling of *engrossment* in the needs of another person and by acting to care for that person.
- For caring to happen, it is necessary for the cared for person to be receptive.
- We can “care about” everyone, but we cannot “care for” everyone. Hence, it is morally acceptable (and necessary) to only care for a small number of individuals

关怀不是单向的，而是一个互动关系（relationship）；
被关怀者（cared-for）也需要愿意接受关怀；
否则，这种“照顾”可能变成控制、干预甚至伤害。

Care about	一种广泛的同情或关心态度（例如“关心世界和平”）	抽象、普遍
Care for	实际照顾、投入时间和情感、建立关系	具体、个人化

- Since caring involves the specific situation of the specific individual, it does not lead to universal moral rules. 每一次关怀行为都与特定的人、特定情境有关
 - Focus on specific cases and analogy, **not** reasoning.



Caring by Computer Professionals

- It's not possible to “care for” all (hundreds or millions of) users of a software application; but, software *can* help people.
- Computer professionals can ask important questions related to the ethics of caring:-
 - ☐ Does the action enhance the ideal of myself as one-caring, or detract from it?
 - ☐ Will the product that I am creating allow its users to enhance their ideals of *themselves* as one-caring, or detract from it?
 - ☐ Is the person I am trying to help actually receptive to my help?
 - ☐ Is my mental attitude one of engrossment in the care of users needs, or am I mentally distant?

关怀伦理在计算机行业中的意义是：

用技术表达同理心，用设计体现人文精神。真正“有道德的程序员”，不仅写出高效的代码，更能以关怀之心理解并尊重用户的需求。



Caring by Computer Professionals

Examples to think about:

- If designing a GUI interface, do you give attention to making the interface beautiful and easy-to-use?
- In particular, do you feel motivated to provide accessibility features (e.g. changing font size and colour).
- If the system is a health system, does it assist and encourage health workers as carers for their patients?

层次	实现方式	道德意义
直接关怀 (Direct Care)	设计易用、美观的界面	尊重用户体验
间接关怀 (Indirect Care)	提供无障碍支持	包容弱势群体
系统性关怀 (Systemic Care)	帮助他人去关怀他人 (如医疗系统)	扩大社会影响力

关怀伦理在计算机设计中的核心是：
“让技术不只是‘能用’，而是‘贴心’。”

设计应当以人性为中心，用科技去传递同理心与尊重。



Workshop w/c 7th October

- Apply Theories of Ethics to scenarios.

